MSG No. 194.270 $P6_3/mm'c'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3m'`.

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,3,5,7,9,11,13,14,15,19,20,21]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,4,6,8,10,12,16,17,18,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,3,5,8,10,12,16,17,18,19,20,21]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,4,6,7,9,11,13,14,15,22,23,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,3,5,8,10,12,16,17,18,19,20,21]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[2,4,6,7,9,11,13,14,15,22,23,24]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[1,3,5,8,10,12,16,17,18,19,20,21]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[2,4,6,7,9,11,13,14,15,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m'`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,3,5,19,20,21]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,4,6,22,23,24]`
3	$[0,\ 0,\ -z]$	`[7,9,11,13,14,15]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[8,10,12,16,17,18]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m'`.

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,3,5,19,20,21]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	`[2,4,6,22,23,24]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,9,11,13,14,15]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	`[8,10,12,16,17,18]`

Table 7: Wyckoff site: 6g, site symmetry: .2'/m'.

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,7,13,19]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,8,16,22]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,9,14,20]
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[4,10,17,23]
5	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[5,11,15,21]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,12,18,24]

Table 8: Wyckoff site: 6h, site symmetry: mm'2'

No.	position	mapping
1	$\left[x, 2x, \frac{1}{4}\right]$	[1,10,17,19]
2	$\left[-x, x, \frac{3}{4}\right]$	[2,11,15,22]
3	$\left[-2x, -x, \frac{1}{4}\right]$	[3,12,18,20]
4	$\left[-x, -2x, \frac{3}{4}\right]$	[4,7,13,23]
5	$\left[x, -x, \frac{1}{4}\right]$	[5,8,16,21]
6	$\left[2x, x, \frac{3}{4}\right]$	[6,9,14,24]

Table 9: Wyckoff site: 12i, site symmetry: .2'.

No.	position	mapping
1	$[x, 0, 0]$	[1,13]
2	$\left[x, x, \frac{1}{2}\right]$	[2,16]
3	$[0, x, 0]$	[3,14]
4	$\left[-x, 0, \frac{1}{2}\right]$	[4,17]
5	$[-x, -x, 0]$	[5,15]
6	$\left[0, -x, \frac{1}{2}\right]$	[6,18]
7	$[-x, 0, 0]$	[7,19]
8	$\left[-x, -x, \frac{1}{2}\right]$	[8,22]
9	$[0, -x, 0]$	[9,20]
10	$\left[x, 0, \frac{1}{2}\right]$	[10,23]
11	$[x, x, 0]$	[11,21]
12	$\left[0, x, \frac{1}{2}\right]$	[12,24]

Table 10: Wyckoff site: 12j, site symmetry: m..

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	[1,10]
2	$\left[x-y, x, \frac{3}{4}\right]$	[2,11]
3	$\left[-y, x-y, \frac{1}{4}\right]$	[3,12]
4	$\left[-x, -y, \frac{3}{4}\right]$	[4,7]
5	$\left[-x+y, -x, \frac{1}{4}\right]$	[5,8]
6	$\left[y, -x+y, \frac{3}{4}\right]$	[6,9]
7	$\left[x-y, -y, \frac{3}{4}\right]$	[13,23]

continued ...

Table 10

No.	position	mapping
8	$\left[y,\ x,\ \frac{3}{4}\right]$	[14,24]
9	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[15,22]
10	$\left[x,\ x-y,\ \frac{1}{4}\right]$	[16,21]
11	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	[17,19]
12	$\left[-y,\ -x,\ \frac{1}{4}\right]$	[18,20]

Table 11: Wyckoff site: 12k, site symmetry: .m'.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,19]
2	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[2,22]
3	$[-2x,\ -x,\ z]$	[3,20]
4	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	[4,23]
5	$[x,\ -x,\ z]$	[5,21]
6	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	[6,24]
7	$[-x,\ -2x,\ -z]$	[7,13]
8	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	[8,16]
9	$[2x,\ x,\ -z]$	[9,14]
10	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	[10,17]
11	$[-x,\ x,\ -z]$	[11,15]
12	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	[12,18]

Table 12: Wyckoff site: 24l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$[-y,\ x-y,\ z]$	[3]
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[4]
5	$[-x+y,\ -x,\ z]$	[5]
6	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	[6]
7	$[-x,\ -y,\ -z]$	[7]
8	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[8]
9	$[y,\ -x+y,\ -z]$	[9]
10	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[10]
11	$[x-y,\ x,\ -z]$	[11]
12	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[12]
13	$[x-y,\ -y,\ -z]$	[13]
14	$[y,\ x,\ -z]$	[14]
15	$[-x,\ -x+y,\ -z]$	[15]
16	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[17]
18	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[18]
19	$[-x+y,\ y,\ z]$	[19]

continued ...

Table 12

No.	position	mapping
20	$[-y,\, -x,\, z]$	[20]
21	$[x,\, x-y,\, z]$	[21]
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[24]